TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
ax + by + c = 0
ax² + by + c = 0
TATA
CONSULTANCY
SERVICES
EQUATIONS
7:49:03 / 11:50:56
TCS Linear Equation
Linear Equation

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Q1. In a competitive exam, 5 marks are awarded for every correct answer and for every wrong answer, 2 marks are deducted. Sathwik scores 32 marks in the examination. If 4 marks had been awarded for each correct answer and 1 mark had been deducted for each incorrect answer, Sathwik would have scored 34 marks. If Sathwik attempted all the questions, how many questions were there in the test ?

A. 14
B. 26 ✓
12
20

Total answers correct = c
Total answers wrong = w
} total questions = c+w

(I) $5c - 2w = 32$ — ①

(II) $4c - w = 34$ — ②

$5c - 2w = 32$
$8c - 2w = 68$

$-5c + 2w = -32$
$8c - 2w = 68$
$3c = 36$

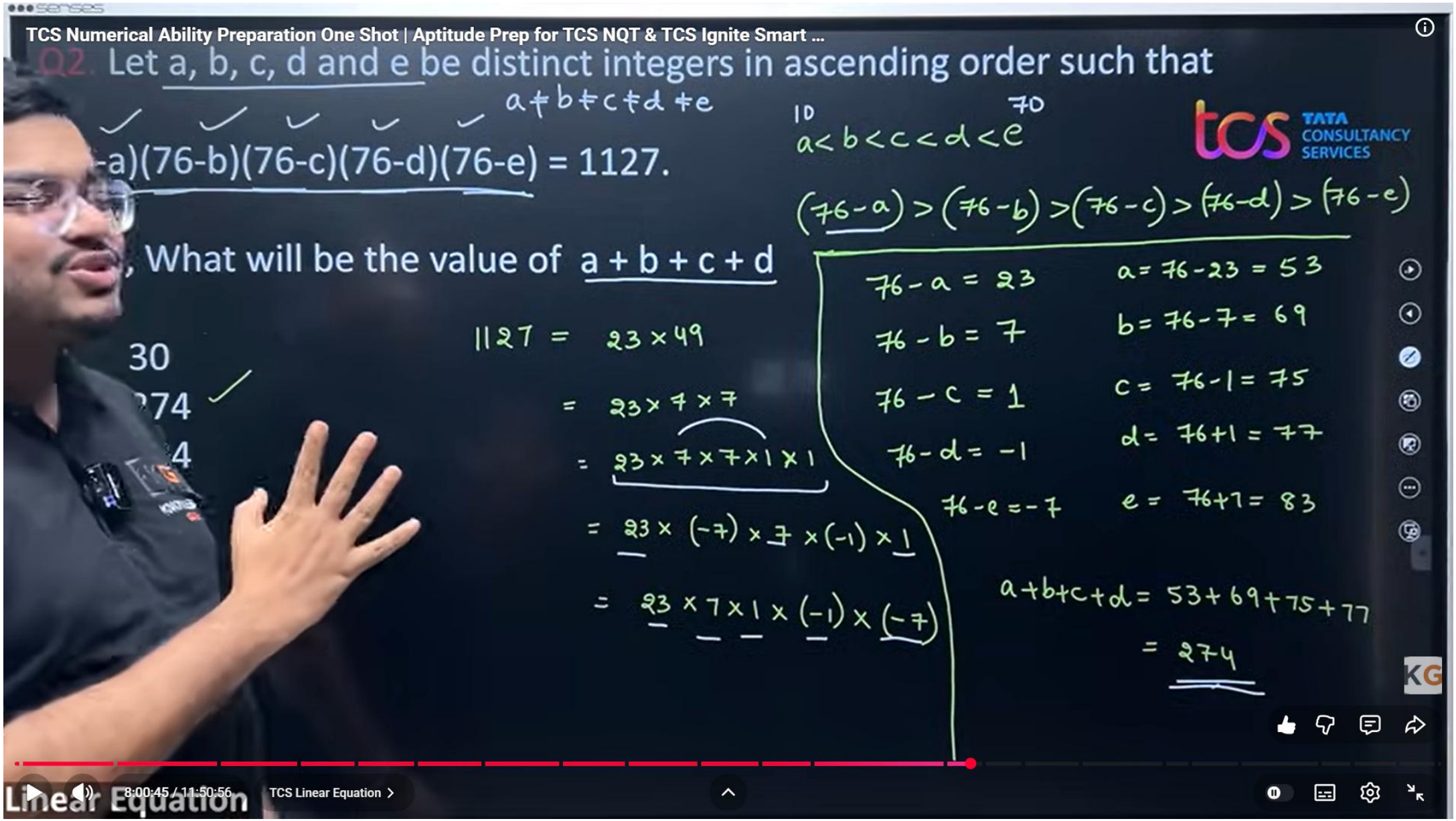
$c = \frac{36}{3} = 12$

$(4 \times 12) - w = 34$

$48 - w = 34 \Rightarrow w = 48 - 34$
$w = 14$

$c + w = 12 + 14$
$= 26$

TATA CONSULTANCY SERVICES
Linear Equation
7:51:04 / 11:50:56
TCS Linear Equation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Q2. Let a, b, c, d and e be distinct integers in ascending order such that
a≠b≠c≠d≠e
-a)(76-b)(76-c)(76-d)(76-e) = 1127.
, What will be the value of a + b + c + d
30
274
4
1127 = 23 × 49
= 23 × 7 × 7
= 23 × 7 × 7 × 1 × 1
= 23 × (-7) × 7 × (-1) × 1
= 23 × 7 × 1 × (-1) × (-7)
10
a < b < c < d < e
70
(76-a) > (76-b) > (76-c) > (76-d) > (76-e)
76 - a = 23
76 - b = 7
76 - c = 1
76 - d = -1
76 - e = -7
a = 76 - 23 = 53
b = 76 - 7 = 69
c = 76 - 1 = 75
d = 76 + 1 = 77
e = 76 + 7 = 83
a+b+c+d = 53+69+75+77
= 274
TATA CONSULTANCY SERVICES
Linear Equation
8:00:45 / 11:50:56
TCS Linear Equation

Q3. Assume that f(1)=0 and f(m+n) = f(m) + f(n) + 4(9mn-1).
For all natural numbers (Integers>0) m and n. What is the value of

A. 5436
B. 4831
C. 5508
D. 4832 ✓

$f(1) \Rightarrow m=1 \quad n=1 \Rightarrow f(2)$

$f(2) \Rightarrow m=2 \quad n=2 \Rightarrow f(4)$

$f(4) \Rightarrow m=4 \quad n=4 \Rightarrow f(8)$

$f(8) \Rightarrow m=8 \quad n=8 \Rightarrow f(16)$

$f(16) \Rightarrow m=16 \quad n=1 \Rightarrow f(17)$ ✓

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